The 6 Parent Functions Name: __________________

Pre-Calculus | Guided notes — we fill these in together

Vocabulary

Parent function: the ___________ member of a family of functions — no shifts, stretches, or reflections applied.

Domain: the set of all allowed _________ (x) values a function can use.

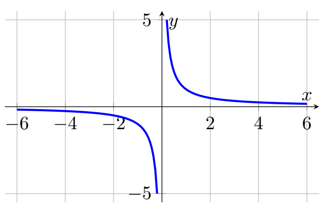
Range: the set of all __________ (y) values a function actually produces.

We do together. Identify the parent function of $f(x) = \dfrac{1}{x}$ and give its domain and range.

Step 1. The rule is a fraction with x in the denominator, so this is the ____________ parent function.

Step 2. A denominator can never equal 0, so x cannot be 0: domain is __________.

Step 3. Since $y = 1/x$ never produces an output of exactly 0 either, the range is __________.



$y = 1/x$ — two branches, never touching either axis (domain $x \neq 0$, range $y \neq 0$).

You Try. Solve each. Show every step, then name the answer.

a) Identify the parent function of $f(x) = x^3$ and give its domain and range.

b) Identify the parent function of $f(x) = |x|$ and give its range.

Summary (fill in): To identify a parent function: match the _________ to one of the 6 known shapes, then check for _____________ caused by a root or a denominator.

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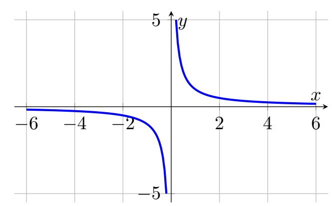

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